MPG No. 11.3.37 $4'/m$ [Type III, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -M_{xxz} & 0 \\ M_{xxz} & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_{z}^{\beta(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yzx} = \mathbb{T}_{v}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{T}_{xy}^{(1)}$$
$$2M_{yzx} = 2\mathbb{T}_{v}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ M_{yzy} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{v}^{(2)}$$
$$M_{yzy} = \mathbb{T}_{xy}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).